Tanisha Krishnaraj

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Education

Boston University Boston, MA

Bachelor of Science in Applied & Mathematics Data Science (GPA: 3.84) *Expected May 2028*

- **Honors/Awards:** Dean's List (All Semesters)
- **Relevant Coursework:** Computational Linear Algebra, Probability Theory, Stochastic Processes, Algorithms

Technical Skills

Languages: Julia, Rust, C, Python (NumPy, Pandas, Scikit-learn, PyTorch), R, SQL
Mathematical Modeling: Numerical Optimization, Spectral Methods, Stochastic Modeling
Systems & Infrastructure: Linux (Bash), OpenSSL, TCP/IP, Encryption, EDR (CrowdStrike Falcon, Carbon Black)

Research Experience

Directed Reading Program (DRP) | Boston University Boston, MA

Research Fellow — Algebraic Graph Theory & Spectral Methods *February 2026 – May 2026*

- Studied network modeling through graph-theoretic representations under PhD supervision, investigating the spectral properties and eigenvalue distributions of adjacency and Laplacian matrices.
- Explored Godsil and Royle's Algebraic Graph Theory, including graph coloring, combinatorial counting, and structural invariants; met regularly with a mentor to present findings and incorporate feedback.

Technical Projects

Loss Landscape & Gradient Descent Analysis | *Python, NumPy, Matplotlib* October 2026

- Implemented and analyzed gradient descent for housing price regression, reducing Hessian condition number from 5.88×10^{1} to 20.88 through feature scaling.
- Derived optimal step size analytically, accelerating gradient descent convergence by 28× (from 5,000 down to 176 iterations) and matching closed-form regression coefficients within 3.84×10^{-6}.

Work Experience

Information Security Compliance | Boston University Boston, MA

Security Assistant *March 2026 – Present*

- Managed endpoint security configuration and system integrity for 200+ research nodes using CrowdStrike Falcon and VMware Carbon Black to prevent unauthorized data exfiltration.
- Engineered automated Bash and Python routines to parse vulnerability reports, streamline ServiceNow tracking, and deploy systematic configuration patches.

Center for Computing & Data Sciences | Boston University Boston, MA

Course Facilitator: Linear Algebra *January 2026 – May 2026*

- Mentored 60+ students on Singular Value Decomposition, Spectral Decomposition, Vector Spaces, and Markov Chains, explaining matrix transformations clearly in writing and in person.
- Organized weekly technical meetings to bridge abstract theory with practical algorithmic implementations of linear algebra.

Leadership Experience

PROMYS Pathways Boston, MA

Mentor *June 2026 – August 2026*

- Mentored a 30-student cohort through intensive number theory and modular arithmetic concepts, using inquiry-based guidance to build mathematical intuition over rote instruction.
- Facilitated weekly collaborative problem-solving sessions, coaching students to discover proofs independently and self-correct their reasoning.